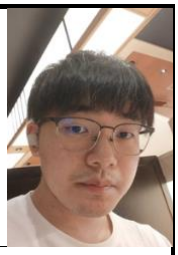
# Curriculum Vitae


| Haocheng Zhu / 朱灏丞 PhD Student | | | |  |
|---|---|---|---|---|
| **Affiliation:** | Failab, Graduate School of Information Sciences, Tohoku University | | | |
| **Location:** | Sendai, Miyagi, Japan | **Email:** | zhu.haocheng.r4@dc.tohoku.ac.jp | |
| **ORCID:** | https://orcid.org/0009-0002-7614-0977 | | | |
| **Personal Website:** | https://zhc-studio.github.io | | | |


## Education

| Period | Institution | Degree / Status |
|---|---|---|
| Apr 2026 – Present | **Tohoku University**<br>Graduate School of Information Sciences<br>Department of System Information Sciences<br>Doctoral Program | **In Progress**<br>Expected Mar 2029 |
| Apr 2024 – Mar 2026 | **Tohoku University**<br>Graduate School of Information Sciences<br>Department of System Information Sciences<br>Master's Program | **Completed**<br>Master of Information Sciences |
| Sep 2018 – Jun 2022 | **Central South University of Forestry and Technology**<br>College of Computer and Information Engineering<br>Software Engineering | **Graduated**<br>Bachelor's Degree |

## Research Experience

### Data Selection for Efficient Supervised Fine-Tuning of Large Language Models

Apr 2025 – Sep 2026 **Failab, Tohoku University**
Advisors: Jun Suzuki, Tianqi Wang

Investigated data selection for supervised fine-tuning (SFT) of large language models under limited training-token budgets. Developed and evaluated a role-separated selection framework combining likelihood-based prioritization, learner-compatibility filtering, and cosine-similarity-based redundancy control to construct effective training subsets.

### Investigating MosaicBERT for Japanese–English Word Alignment

Oct 2024 – Jan 2025 **Failab, Tohoku University**
Advisors: Masaaki Nagata (NTT), Jun Suzuki

Explored the use of MosaicBERT, an encoder-only model, for Japanese–English word alignment. Investigated approaches to improving model performance and the applicability of multilingual BERT-based models to domain-specific translation tasks.

### Learning Rate Selection for Continual Pretraining of Large Language Models

Jun 2024 – Sep 2024 **Failab, Tohoku University**
Advisor: Jun Suzuki

Experimentally investigated the effects of learning-rate selection on continual pretraining using Qwen2. Focused on mitigating catastrophic forgetting while balancing knowledge retention and acquisition of new knowledge.

## Publications & Presentations

**【Peer-Reviewed International Conference Papers】**

Haocheng Zhu, Tianqi Wang, Jun Suzuki.
"Beyond Ranking: Role-Specific Signals for Token-Budgeted SFT Data Selection."
**Findings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP 2026 Findings)**, 2026.

**【Domestic Conference Papers】**

Haocheng Zhu, Tianqi Wang, Jun Suzuki.
「品質と網羅性に基づくデータ選択による言語モデルの効率的学習」
[Efficient Training of Language Models through Data Selection Based on Quality and Coverage]
**Proceedings of the 32nd Annual Meeting of the Association for Natural Language Processing (NLP 2026)**, pp. 3264–3269, 2026. (in Japanese)

**【Conference & Workshop Presentations】**

Haocheng Zhu et al.
"A Study on Learning Rate Selection in Continual Pretraining of Large Language Models."
**19th Symposium of Young Researchers in NLP (YANS 2024)**, Poster Presentation, Osaka, Japan, September 2024. (in Japanese)

## Theses

**【Master's Thesis】**

"Efficient Training of Language Models via Data Selection based on Quality and Coverage"
Master's Thesis, Graduate School of Information Sciences, Tohoku University, March 2026.
Advisor: Jun Suzuki

**【Undergraduate Capstone Project】**

"Design and Implementation of a Game Based on EEG Interaction"
Graduation Project, Central South University of Forestry and Technology, June 2022. (in Chinese)
Advisor: Hui Huang

## Languages

| | |
|---|---|
| **Japanese** | JLPT N1　106 points |
| **English** | TOEFL iBT　82　points |
| **Chinese** | Native |

## Honors & Awards

| | |
|---|---|
| 2024 | YANS Organizing Committee Special Award (Senryu Hackathon)<br>19th YANS Symposium (YANS 2024) |

## Fellowships & Funding

| **Period** | **Program / Support** |
|---|---|
| Apr 2026 – Present | Tohoku University Advanced Graduate School LEAP Program<br>Incentive Stipend Recipient |
| Apr 2025 – Present | Graduate Program in Data Science (GP-DS), Tohoku University<br>Program Member |

## Project Experience

| Period | Project |
|---|---|
| Nov 2025 – Mar 2026 | **Telecom-Specific LLM Development Project**<br>Failab, Tohoku University × SoftBank Corp<br>Contributed to the development of a telecom-specific LLM for the GSMA Open-Telco LLM Benchmarks, primarily focusing on continual pretraining (CPT) data preparation. Selected telecom-domain data from large-scale Web corpora, including Dolma3 and FineWeb, using NVIDIA Domain Filter and a ModernBERT-based classifier, constructing a dataset of approximately 19.48 million samples (67 GB). Gained experience with Singularity-based GPU environments, GitHub version control, and university–industry collaboration. SoftBank's LTM ranked third overall in the GSMA Open-Telco LLM Benchmarks in March 2026. |

## Internship Experience

| Period | Position / Company |
|---|---|
| Mar 2022 – Jun 2022 | Game Tester (Intern)<br>Hunan Caohua Interactive Network Technology Co., Ltd. |

## Technical Skills

**【Programming & Machine Learning】**
Data preprocessing, ML/LLM training, inference, evaluation, and analysis using Python.
Model and dataset management, and LLM training and evaluation pipelines using Hugging Face.

**【HPC & Computing Environments】**
Experience with GENKAI, ABCI, mdx, laboratory GPU servers, and Hokkaido University GC2.
Container environment setup and management using Singularity.
Python virtual environment and dependency management using uv.
Linux server operations, job execution, and experimental environment setup.

**【Development & Experiment Management】**
Experiment tracking using Weights & Biases (W&B).
Version control and collaborative development using Git/GitHub.